This is a preview of "2020 ASHRAE Handbook...". Click here to purchase the full version from the ANSI store.

2020 ASHRAE HANDBOOK

HVAC SYSTEMS AND EQUIPMENT

SI Edition

Supported by ASHRAE Research

This is a preview of "2020 ASHRAE Handbook...". Click here to purchase the full version from the ANSI store.

2020 ASHRAE® HANDBOOK

Heating, Ventilating, and Air-Conditioning SYSTEMS AND EQUIPMENT

SI Edition

ASHRAE, 1791 Tullie Circle, N.E., Atlanta, GA 30329
www.ashrae.org

This is a preview of "2020 ASHRAE Handbook...". Click here to purchase the full version from the ANSI store.

© 2020 ASHRAE. All rights reserved.

DEDICATED TO THE ADVANCEMENT OF

THE PROFESSION AND ITS ALLIED INDUSTRIES

No part of this publication may be reproduced without permission in writing from ASHRAE, except by a reviewer who may quote brief passages or reproduce illustrations in a review with appropriate credit; nor may any part of this book be reproduced, stored in a retrieval system, or transmitted in any way or by any means—electronic, photocopying, recording, or other—without permission in writing from ASHRAE. Requests for permission should be submitted at www.ashrae.org/permissions.

Volunteer members of ASHRAE Technical Committees and others compiled the information in this handbook, and it is generally reviewed and updated every four years. Comments, criticisms, and suggestions regarding the subject matter are invited. Any errors or omissions in the data should be brought to the attention of the Editor. Additions and corrections to Handbook volumes in print will be published in the Handbook published the year following their verification and, as soon as verified, on the ASHRAE Internet website.

DISCLAIMER

ASHRAE has compiled this publication with care, but ASHRAE has not investigated, and ASHRAE expressly disclaims any duty to investigate, any product, service, process, procedure, design, or the like that may be described herein. The appearance of any technical data or editorial material in this publication does not constitute endorsement, warranty, or guaranty by ASHRAE of any product, service, process, procedure, design, or the like. ASHRAE does not warrant that the information in this publication is free of errors. The entire risk of the use of any information in this publication is assumed by the user.

ISBN 978-1-947192-53-9
ISSN 1930-7705

The paper for this book is both acid- and elemental-chlorine-free and was manufactured with pulp obtained from sources using sustainable forestry practices.

This is a preview of "2020 ASHRAE Handbook...". Click here to purchase the full version from the ANSI store.

CONTENTS

Contributors

ASHRAE Technical Committees, Task Groups, and Technical Resource Groups

ASHRAE Research: Improving the Quality of Life

Preface

AIR-CONDITIONING AND HEATING SYSTEMS

Chapter
1. **HVAC System Analysis and Selection** (TC 9.1, Large Building Air-Conditioning Systems)
2. **Decentralized Cooling and Heating** (TC 9.1)
3. **Central Cooling and Heating Plants** (TC 9.1)
4. **Air Handling and Distribution** (TC 9.1)
5. **In-Room Terminal Systems** (TC 9.1)
6. **Radiant Heating and Cooling** (TC 6.5, Radiant Heating and Cooling)
7. **Combined Heat and Power Systems** (TC 1.10, Cogeneration Systems)
8. **Combustion Turbine Inlet Cooling** (TC 1.10)
9. **Applied Heat Pump and Heat Recovery Systems** (TC 6.8, Geothermal Heat Pump and Energy Recovery Applications)
10. **Small Forced-Air Heating and Cooling Systems** (TC 6.3, Central Forced Air Heating and Cooling Systems)
11. **Steam Systems** (TC 6.1, Hydronic and Steam Equipment and Systems)
12. **District Heating and Cooling** (TC 6.2, District Energy)
13. **Hydronic Heating and Cooling** (TC 6.1)
14. **Condenser Water Systems** (TC 6.1)
15. **Medium- and High-Temperature Water Heating** (TC 6.1)
16. **Infrared Radiant Heating** (TC 6.5)
17. **Ultraviolet Lamp Systems** (TC 2.9, Ultraviolet Air and Surface Treatment)
18. **Variable Refrigerant Flow** [TC 8.7, Variable Refrigerant Flow (VRF)]

AIR-HANDLING EQUIPMENT AND COMPONENTS

Chapter
19. **Duct Construction** (TC 5.2, Duct Design)
20. **Room Air Distribution Equipment** (TC 5.3, Room Air Distribution)
21. **Fans** (TC 5.1, Fans)
22. **Humidifiers** (TC 5.11, Humidifying Equipment)
23. **Air-Cooling and Dehumidifying Coils** (TC 8.4, Air-to-Refrigerant Heat Transfer Equipment)
24. **Desiccant Dehumidification and Pressure-Drying Equipment** (TC 8.12, Desiccant Dehumidification Equipment and Components)
25. **Mechanical Dehumidifiers and Related Components** (TC 8.10, Mechanical Dehumidification Equipment and Heat Pipes)
26. **Air-to-Air Energy Recovery Equipment** (TC 5.5, Air-to-Air Energy Recovery)
27. **Air-Heating Coils** (TC 8.4)
28. **Unit Ventilators, Unit Heaters, and Makeup Air Units** (TC 6.1 and TC 5.8, Industrial Ventilation)
29. **Air Cleaners for Particulate Contaminants** (TC 2.4, Particulate Air Contaminants and Particulate Contaminant Removal Equipment)
30. **Industrial Gas Cleaning and Air Pollution Control** [TC 5.4, Industrial Process Air Cleaning (Air Pollution Control)]

This is a preview of "2020 ASHRAE Handbook...". Click here to purchase the full version from the ANSI store.

HEATING EQUIPMENT AND COMPONENTS

Chapter 31. **Automatic Fuel-Burning Systems** (TC 6.10, Fuels and Combustion)
32. **Boilers** (TC 6.1)
33. **Furnaces** (TC 6.3)
34. **Residential In-Space Heating Equipment** (TC 6.5)
35. **Chimney, Vent, and Fireplace Systems** (TC 6.10)
36. **Hydronic Heat-Distributing Units and Radiators** (TC 6.1)
37. **Solar Energy Equipment** (TC 6.7, Solar Energy Utilization)

COOLING EQUIPMENT AND COMPONENTS

Chapter 38. **Compressors** (TC 8.1, Positive Displacement Compressors, and TC 8.2, Centrifugal Machines)
39. **Condensers** (TC 8.4, TC 8.5, Liquid-to-Refrigerant Heat Exchangers, and TC 8.6, Cooling Towers and Evaporative Condensers)
40. **Cooling Towers** (TC 8.6)
41. **Evaporative Air-Cooling Equipment** (TC 5.7, Evaporative Cooling)
42. **Liquid Coolers** (TC 8.5)
43. **Liquid-Chilling Systems** (TC 8.1 and TC 8.2)

GENERAL COMPONENTS

44. **Centrifugal Pumps** (TC 6.1)
45. **Motors, Motor Controls, and Variable-Frequency Drives** (TC 1.11, Electric Motors and Motor Control)
46. **Pipes, Tubes, and Fittings** (TC 6.1)
47. **Valves** (TC 6.1)
48. **Heat Exchangers** (TC 6.1)

PACKAGED, UNITARY, AND SPLIT-SYSTEM EQUIPMENT

Chapter 49. **Unitary Air Conditioners and Heat Pumps** (TC 8.11, Unitary and Room Air Conditioners and Heat Pumps)
50. **Room Air Conditioners and Packaged Terminal Air Conditioners** (TC 8.11)

GENERAL

51. **Thermal Storage** (TC 6.9, Thermal Storage)
52. **Dedicated Outdoor Air Systems** (TC 8.10
53. **Codes and Standards**

Additions and Corrections

Index

Composite index to the 2017 Fundamentals, 2018 Refrigeration, 2019 HVAC Applications, and 2020 HVAC Systems and Equipment volumes

Comment Pages

This is a preview of "2020 ASHRAE Handbook...". Click here to purchase the full version from the ANSI store.

CONTRIBUTORS

In addition to the Technical Committees, the following individuals contributed significantly to this volume. The appropriate chapter numbers follow each contributor's name.

Charlie Henck, PE, FASHRAE (2)
Whitman, Requardt and Associates, LLP

Jay Eldridge, LEED AP (3)
Daikin Applied

Phil Trafton, FASHRAE (5)
Donald F. Dickerson, Assoc.

Rex Scare, PE (11)
Armstrong International

Eric J. Rosenberg (14)
Grumman/Butkus Associates

Steve Tredinnick, PE, CEM (14, 15)
Burns & McDonnell, Inc.

Jaak Geboers (17)

Signify Netherlands BV

Stephen B. Martin, PhD, PE (17)
CDC, NIOSH

Dean Saputa (17)
UVResources

Kevin Gepke (20)
DuctSox Corp.

Ryan Johnson (20)
Price Industries

Robert Persechini (20)
NV5

Joseph Brooks (21)
Air Movement & Control Association

Armin Hauer (21)
Ebm-Papst Inc.

Tim Mathson (21)
Air Movement & Control Association

Zhiping Wang, PhD (21)
Morrison Products, Inc.

Craig Wray (21)

Brandon Boor (29)
Purdue University

Zied Driss (29)
Camfil USA, Inc.

Dan Haas (29)
Parker Hannifin HVAC

Carolyn M. L. Kerr, PhD (29)

Brian Krafthefer (29)
BCK Consulting

Don Thornburg (29)
Camfil USA, Inc.

Kwang Woo Kim, Arch.D. (34)
Seoul National University

Aykut Yilmaz (34)
AHRI

Thomas Neill (36)
Mestek

Mark Hertel (37)
Inter-Island Solar Supply

Konstantinos Kapsis, PhD (37)
Natural Resources Canada

Scott Schuetter (37)
Slipstream

Lee Shaver (37)
Slipstream

Dwayne Johnson (38)

Trane

Drew Turner (38)
Danfoss

Trent Hunt (38)
Mechanical Products Intermountain, Inc.

Michael H. Zamalis, PE (38)
Johnson Controls

Adam Radford (40)
Evapco, Inc.

Jian Xu (40)
BAC

Jason Atkisson (44)
Affiliated Engineers, Inc.

David Lee (44)
Armstrong Fluid Technology

Thomas Lowery (45)
Schneider Electric

Tricia M. Bruenn (46)

Robert Walker (46)
Belimo

Scott Fisher (47)
Hurley Engineering

Robert Berry, MSME (48)
Unico, Inc.

Ted Duffy, PE (48)
LAARS Heating Systems

Henry (Skip) Ernst (48)
Daikin

Kevin B. Mercer, PE (48)
Carrier Corp.

Raymond Rite, PhD (48)
Ingersoll Rand

Guy Frankenfield (50)
DN Tanks

Kenneth M. Fulk, PE (50)
Reed, Wells, Benson and Company

John Lau (50)

BAC Jessup

Daniel Pyewell, PE (50)
Tanco Engineering, Inc.

This is a preview of "2020 ASHRAE Handbook...". Click here to purchase the full version from the ANSI store.

ASHRAE HANDBOOK COMMITTEE

Suzanne LeViseur, Chair

2020 HVAC Systems and Equipment Volume Subcommittee: **Michael P. Patton,** Chair

Caroline C. Calloway **Nicolas Lemire** **Prakash R. Dhamshala** **Florentino Roson Rodriguez** **Steven C. Sill**

ASHRAE HANDBOOK STAFF

Mark S. Owen, Publisher
Director of Publications and Education

Heather E. Kennedy, Editor

Hayden Spiess, Editorial Assistant

Nancy F. Thysell, Typographer/Page Designer

David Soltis, Group Manager, and **Jayne E. Jackson,** Publication Traffic Administrator Publishing Services

This is a preview of "2020 ASHRAE Handbook...". Click here to purchase the full version from the ANSI store.

ASHRAE TECHNICAL COMMITTEES, TASK GROUPS, AND TECHNICAL RESOURCE GROUPS

SECTION 1.0—FUNDAMENTALS AND GENERAL

1.1 Thermodynamics and Psychrometrics
1.2 Instruments and Measurements
1.3 Heat Transfer and Fluid Flow
1.4 Control Theory and Application
1.5 Computer Applications
1.6 Terminology
1.7 Business, Management & General Legal Education
1.8 Mechanical Systems Insulation
1.9 Electrical Systems
1.10 Cogeneration Systems
1.11 Electric Motors and Motor Control
1.12 Moisture Management in Buildings
1.13 Optimization

SECTION 2.0—ENVIRONMENTAL QUALITY

2.1 Physiology and Human Environment
2.2 Plant and Animal Environment
2.3 Gaseous Air Contaminants and Gas Contaminant Removal Equipment
2.4 Particulate Air Contaminants and Particulate Contaminant Removal Equipment
2.5 Global Climate Change
2.6 Sound and Vibration
2.7 Seismic and Wind Resistant Design
2.8 Building Environmental Impacts and Sustainability
2.9 Ultraviolet Air and Surface Treatment
TG2 Heating Ventilation and Air-Conditioning Security (HVAC)

SECTION 3.0—MATERIALS AND PROCESSES

3.1 Refrigerants and Secondary Coolants
3.2 Refrigerant System Chemistry
3.3 Refrigerant Contaminant Control
3.4 Lubrication
3.6 Water Treatment
3.8 Refrigerant Containment

SECTION 4.0—LOAD CALCULATIONS AND ENERGY REQUIREMENTS

4.1 Load Calculation Data and Procedures
4.2 Climatic Information
4.3 Ventilation Requirements and Infiltration
4.4 Building Materials and Building Envelope Performance
4.5 Fenestration
4.7 Energy Calculations
4.10 Indoor Environmental Modeling
TRG4 Indoor Air Quality Procedure Development

SECTION 5.0—VENTILATION AND AIR DISTRIBUTION

5.1 Fans
5.2 Duct Design
5.3 Room Air Distribution
5.4 Industrial Process Air Cleaning (Air Pollution Control)
5.5 Air-to-Air Energy Recovery
5.6 Control of Fire and Smoke
5.7 Evaporative Cooling
5.8 Industrial Ventilation
5.9 Enclosed Vehicular Facilities
5.10 Kitchen Ventilation
5.11 Humidifying Equipment

SECTION 6.0—HEATING EQUIPMENT, HEATING AND COOLING SYSTEMS AND APPLICATIONS

6.1 Hydronic and Steam Equipment and Systems
6.2 District Energy
6.3 Central Forced Air Heating and Cooling Systems
6.5 Radiant Heating and Cooling
6.6 Service Water Heating Systems
6.7 Solar Energy Utilization
6.8 Geothermal Heat Pump and Energy Recovery Applications
6.9 Thermal Storage
6.10 Fuels and Combustion

SECTION 7.0—BUILDING PERFORMANCE

7.1 Integrated Building Design
7.2 HVAC&R Construction & Design Build Technologies
7.3 Operation and Maintenance Management
7.4 Exergy Analysis for Sustainable Buildings (EXER)
7.5 Smart Building Systems
7.6 Building Energy Performance
7.7 Testing and Balancing
7.8 Owning and Operating Costs
7.9 Building Commissioning

SECTION 8.0—AIR-CONDITIONING AND REFRIGERATION SYSTEM COMPONENTS

8.1 Positive Displacement Compressors
8.2 Centrifugal Machines
8.3 Absorption and Heat Operated Machines
8.4 Air-to-Refrigerant Heat Transfer Equipment
8.5 Liquid-to-Refrigerant Heat Exchangers
8.6 Cooling Towers and Evaporative Condensers
8.7 Variable Refrigerant Flow (VRF)
8.8 Refrigerant System Controls and Accessories
8.9 Residential Refrigerators and Food Freezers
8.10 Mechanical Dehumidification Equipment and Heat Pipes
8.11 Unitary and Room Air Conditioners and Heat Pumps
8.12 Desiccant Dehumidification Equipment and Components

SECTION 9.0—BUILDING APPLICATIONS

9.1 Large Building Air-Conditioning Systems
9.2 Industrial Air Conditioning
9.3 Transportation Air Conditioning
9.4 Justice Facilities
9.6 Healthcare Facilities
9.7 Educational Facilities
9.8 Large Building Air-Conditioning Applications
9.9 Mission Critical Facilities, Data Centers, Technology Spaces and Electronic Equipment
9.10 Laboratory Systems
9.11 Clean Spaces
9.12 Tall Buildings

SECTION 10.0—REFRIGERATION SYSTEMS

10.1 Custom Engineered Refrigeration Systems
10.2 Automatic Icemaking Plants and Skating Rinks
10.3 Refrigerant Piping, Controls, and Accessories
10.5 Refrigerated Processing and Storage
10.6 Transport Refrigeration
10.7 Commercial Food and Beverage Refrigeration Equipment
10.8 Refrigeration Load Calculations

SECTION MTG—MULTIDISCIPLINARY TASK GROUPS

MTG.ASEC Avoided Sources Energy Consumption Due to Waste Heat Recovery and Heat Pump Technologies
MTG.BD Building Dampness
MTG.BIM Building Information Modeling
MTG.ET Energy Targets
MTG.HCDG Hot Climate Design Guide
MTG.IAST Impact of ASHRAE Standards and Technology on Energy Savings/Performance
MTG.ISPAQE Indoor Swimming Pool Air Quality and Evaporation
MTG.LowGWP Lower Global Warming Potential Alternative Refrigerants
MTG.O&MEE Operations and Maintenance Activities That Impact Energy Efficiency
MTG.OBB Occupant Behavior in Buildings

This is a preview of "2020 ASHRAE Handbook...". Click here to purchase the full version from the ANSI store.

ASHRAE Research: Improving the Quality of Life

ASHRAE is the world's foremost technical society in the fields of heating, ventilation, air conditioning, and refrigeration. Its members worldwide are individuals who share ideas, identify needs, support research, and write the industry's standards for testing and practice. The result is that engineers are better able to keep indoor environments safe and productive while protecting and preserving the outdoors for generations to come.

One of the ways that ASHRAE supports its members' and industry's need for information is through ASHRAE Research. Thousands of individuals and companies support ASHRAE Research annually, enabling ASHRAE to report new data about material properties and building physics and to promote the application of innovative technologies.

Chapters in the ASHRAE Handbook are updated through the experience of members of ASHRAE Technical Committees and through results of ASHRAE Research reported at ASHRAE conferences and published in ASHRAE special publications, *ASHRAE Transactions*, and ASHRAE's journal of archival research, *Science and Technology for the Built Environment*.

For information about ASHRAE Research or to become a member, contact ASHRAE, 1791 Tullie Circle N.E., Atlanta, GA 30329; telephone: 404-636-8400; www.ashrae.org.

Preface

The 2020 *ASHRAE Handbook—HVAC Systems and Equipment* discusses various systems and the equipment (components or assemblies) they comprise, and describes features and differences. This information helps system designers and operators in selecting and using equipment. ASHRAE Technical Committees in each subject area have reviewed all chapters and revised them as needed for current technology and practice. An accompanying CD-ROM contains all the volume's chapters in both I-P and SI units.

Some of the volume's revisions and additions are as follows:

- Chapter 7, Combined Heat and Power Systems, has a new section on economic evaluation and includes an update on EU Directive 2004/8/EC.
- Chapter 9, Applied Heat Pump and Heat Recovery Systems, has new content on waste heat recovery, district applications, and industrial process heat pumps.
- Chapter 12, District Heating and Cooling, has new content from ASHRAE research project RP-1267 (the new *District Heating Guide* and *District Cooling Guide*).
- Chapter 18, Variable Refrigerant Flow, has new sections on modeling and system commissioning, and an updated system design example.
- Chapter 19, Duct Construction, has extensive revisions on system leakage and air dispersion systems.
- Chapter 20, Room Air Distribution Equipment, has updates for current technology, with new information on specialized components and air curtains.
- Chapter 21, Fans, has new sections on series fan operation and field performance testing plus added content on fan and motor efficiency grades and parallel multiple-fan operation.
- Chapter 24, Desiccant Dehumidification and Pressure-Drying Equipment, has expanded content on applications, air filters, and liquid strainers, plus recommendations from ASHRAE research project RP-1339 on rating equipment at altitude.
- Chapter 25, Mechanical Dehumidifiers and Related Components, has new content on psychrometrics, outdoor air, controls, and industrial dehumidifiers.
- Chapter 26, Air-to-Air Energy Recovery Equipment, has new information on heat pipes and desiccant and heat wheel systems.
- Chapter 28, Unit Ventilators, Unit Heaters, and Makeup Air Units, has revisions on standards, controls, and fan selection for makeup air units.
- Chapter 29, Air Cleaners for Particulate Contaminants, has updates on standards and performance testing.
- Chapter 31, Automatic Fuel-Burning Systems, has added content on pneumatically and electronically linked gas/air ratio burner systems.
- Chapter 33, Furnaces, has updates for current technology and efficiency requirements.
- Chapter 37, Solar Energy Equipment, has new data on worldwide solar technology use, plus an expanded section on photovoltaic equipment.
- Chapter 38, Compressors, has revisions on general theory; screw and scroll compressors; and bearings, including oil-free technologies.
- Chapter 44, Centrifugal Pumps, has new content on vertical, inline, split-coupled pumps; hydronic system pump selection; and differential pressure control.
- Chapter 45, Motors, Motor Controls, and Variable-Frequency Drives, has new content on standards, bearing currents, and permanent-magnet motors.
- Chapter 47, Valves, has new content on control valve sizing; electronic actuators; and ball, butterfly, flow-limiting, and pressure-independent control valves.
- Chapter 49, Unitary Air Conditioners and Heat Pumps, has a new map of U.S. regional appliance efficiency standards.
- Chapter 50, Room Air Conditioners and Packaged Terminal Air Conditioners, has updates for efficiency standards.
- Chapter 52, Dedicated Outdoor Air Systems, is a completely new chapter focusing on DOAS equipment.

This volume is published, as a bound print volume and in electronic format as a PDF download and online, in two editions: one using inch-pound (I-P) units of measurement, the other using the International System of Units (SI).

Corrections to the 2017, 2018, and 2019 Handbook volumes can be found on the ASHRAE website at www.ashrae.org and in the Additions and Corrections section of this volume. Corrections for this volume will be listed in subsequent volumes and on the ASHRAE website.

Reader comments are enthusiastically invited. To suggest improvements for a chapter, **please comment using the form on the ASHRAE website** or, using the cutout page(s) at the end of this volume's index, write to Handbook Editor, ASHRAE, 1791 Tullie Circle N.E., Atlanta, GA 30329, or fax 678-539-2187, or e-mail hkennedy@ashrae.org.

Heather E. Kennedy
Editor